

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Peer Review

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 – Peer Review
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	N.SATHWIK	Reg. No.:	192525077
Section / Batch:	2/2025	Date:	22/07/2026

Code of Conduct

I N.SATHWIK(192525007) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.SATHWIK

Instructions

- This assessment contains peer review with individual presentation. Total: 100 Marks.
- Students should answer technical questions from peers and faculty.
- No negative marking. Unanswered questions carry zero marks.
- Students can use PowerPoint/Whiteboard/Live Coding/Demonstration for presentation.

Section / Topic	Focus	Q Nos	Marks
Linear Data Structure	Linked List, Stack, Queue	1	100
GRAND TOTAL:			100 Marks

COURSE NAME: CSA0309 DATA STRUCTURES

PEER REVIEW PRESENTATION

TOPIC:

**HOW INSTAGRAM FINDS MILLIONS OF ACCOUNTS
WITHIN A SECOND**



GUIDED BY:
DR. UMA PRIYADARSINI

PRESENTED BY:
N. SATHWIK (192525077)



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ABSTRACT

Instagram is able to find millions of accounts in a fraction of a second by using efficient data structures, indexing techniques, caching mechanisms, and distributed search systems at scale. When a user types a query in the search bar, Instagram first processes the input using text normalization and autocomplete suggestions powered by trie data structures for prefix matching. The actual search is performed using highly optimized search indexes such as inverted indexes and hash tables, which allow quick retrieval of relevant user IDs based on usernames, names, hashtags, or keywords. These indexes are stored in distributed databases and sharded across multiple servers to handle massive data and traffic. Frequently accessed data is served from in-memory caches to reduce latency. Instagram also uses ranking algorithms to sort and prioritize the most relevant results based on popularity, recency, and user interactions. This combination of advanced data structures and system design enables Instagram to deliver accurate search results from millions of accounts within milliseconds.

Annexure A - Peer Review

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concept	20	Demonstrates complete understanding of the selected data structure/search algorithm with accurate explanations.	Minor omissions but concepts are mostly correct.	Partial understanding with noticeable conceptual gaps.	Unable to explain the concept correctly.
Demonstration	20	Demonstrates all operations correctly using suitable examples or live coding	Demonstrates most operations correctly with minor errors.	Demonstrates limited operations with several errors.	Demonstration is incorrect or incomplete.
Presentation & Communication Skills	20	Organized, confident, clear explanation with effective visual aids.	Good communication with minor presentation issues.	Moderate communication; lacks organization.	Poor communication & presentation.
Question Handling	30	Answers all technical questions confidently with correct reasoning.	Answers most questions correctly.	Answers some questions with partial understanding.	Unable to answer technical questions.
Time Management & Organization	10	Completes presentation within allotted time with smooth flow.	Minor deviation from time.	Significant time management issues.	Presentation poorly managed.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____